

# Kiljae Lee

Statistics Ph.D. Candidate | Data-Centric AI | Explainable AI | LLM Evaluation | Uncertainty Quantification

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## Education

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|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Ph.D. in Statistics</b><br><i>The Ohio State University</i> , Advisor: Dr. Yuan Zhang, GPA: 4.00/4.00 | Aug 2022 – Expected Jun 2027<br><i>Columbus, OH</i> |
| <b>M.S. in Statistics</b><br><i>Korea University</i> , Advisor: Dr. Taeryon Choi, GPA: 4.44/4.50         | Mar 2020 – Aug 2021<br><i>Seoul, South Korea</i>    |
| <b>B.S. in Statistics</b><br><i>Korea University</i> , GPA: 4.14/4.50                                    | Mar 2014 – Feb 2020<br><i>Seoul, South Korea</i>    |

## Research Signal

- **4 first-author papers at ICML, NeurIPS, and ICLR, including one co-first-author paper**, bringing statistical insight to data valuation, feature attribution, LLM evaluation, and uncertainty quantification in modern AI systems.
- Bridges **statistical theory and practical AI implementation**, translating Shapley-value and uncertainty quantification ideas into scalable methods for evaluating data quality, model behavior, and prediction uncertainty.
- Identifies **failure modes and adversarial vulnerabilities** in widely used data valuation methods, and develops robust frameworks for reliable data contribution measurement.

## Selected Publications

**Generalized Priority-Aware Shapley Value** (NeurIPS 2026; ICML 2026 CTB Workshop; first author)

Incorporates cyclic human preferences into LLM evaluation by modeling structured preferences as directed weighted priority graphs.

**Priority-Aware Shapley Value** (ICML 2026; first author)

Identifies data copying attacks in data valuation and introduces PASV, a robust data valuation framework that accounts for precedence constraints among data providers.

**Faithful Group Shapley Value** (NeurIPS 2025; ICML 2025 DataWorld Workshop; one of two co-first authors)

Identifies dataset splitting attacks in data valuation and proposes FGSV, a faithful data valuation method that defends against strategic manipulation.

**Leave-One-Out Stable Conformal Prediction** (ICLR 2025; first author)

Achieves  $\geq 100\times$  speedups over standard conformal prediction methods by avoiding data splitting and iterative model fitting at the same time.

## Research and Industry Experience

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>Data Scientist Intern</b><br><i>United Airlines</i>                                                                                                                        | May 2026 – August 2026<br><i>Chicago, IL</i> |
| ▪ Built demand forecasting models that use customer search data as a proxy for latent total demand, improving willingness-to-pay estimation beyond observed booking behavior. |                                              |
| <b>Graduate Research Assistant</b><br><i>The Ohio State University</i>                                                                                                        | Aug 2025 – Present<br><i>Columbus, OH</i>    |
| <b>Statistical Consultant</b><br><i>The Ohio State University</i>                                                                                                             | May 2025 – Jul 2025<br><i>Columbus, OH</i>   |

## Additional Recognition and Service

- **ICML 2026 Gold Reviewer** (top 25% of reviewers); reviewer for ICLR (2025), ICML (2025, 2026), NeurIPS (2025, 2026), and AISTATS (2026).
- **Ransom & Marian Whitney Award for Research 2026**, The Ohio State University; one of two awardees, recognizing Ph.D. research independence, creativity, originality, and publication/application potential.
- **Best Paper Award Honorable Mention**, ICML 2025 DataWorld Workshop, for *Faithful Group Shapley Value*; ranked among the top 3 of 55 accepted submissions.

## Technical Skills

**Programming:** Python, R, C++, SQL, Git

**ML stack:** PyTorch, Hugging Face Transformers, scikit-learn, NumPy/SciPy, pandas

**LLM/Generative AI:** Open-source {LLM/Diffusion} {fint-tuning/inference}, quantized models, PEFT, RLHF

**Infrastructure:** SLURM/HPC clusters, GPU computing, SSH/tmux